

CACYBP expression predicts prognosis in hepatocellular carcinoma

A meta-analysis and bioinformatics validation

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Abstract

Background: Hepatocellular carcinoma (HCC) is one of the most prevalent and inflammation-associated cancers. The potential prognostic value of calyculin-binding protein (CACYBP) in HCC remains unclear. We aimed to perform meta-analysis to clarify the association of CACYBP expression with prognosis.

Methods: We searched various databases to collect all eligible studies. The meta-analysis was performed using Review manager V5.4.1. Gene expression profiling interactive analysis 2 (limma) and differential expression sequencing 2 on R 4.3.3 were used to look for differentially expressed genes. Gene-set enrichment analysis using 4 Gene Ontology databases (molecular function; biological process; cellular component; pathways) was performed using TCGAbiolinks on R. Survival analysis on The Cancer Genome Atlas-HCC was done using gene expression profiling interactive analysis 2.

Results: Four studies were retrieved for meta-analysis. Overall survival (OS) (hazard ratio = 1.08; 95% confidence interval: 1.05–1.10), indicated that high CACYBP expression is associated with worse OS. The Cancer Genome Atlas validation consistently indicated that high CACYBP expression correlates with OS (hazard ratio: 1.24, *P*-value: .04), indicating a worse prognosis. RNA sequencing analysis identified 2910 differentially expressed genes between high and low CACYBP expression groups. Gene-set enrichment analysis showed that upregulated genes in high CACYBP expressers relate to mitosis, nuclear division, cell cycle, microtubule cytoskeleton, ion channel activity, eIF2 signaling, and DNA damage responses. Conversely, downregulated genes were related to oxidation–reduction, steroid metabolic process, response to hormone stimulus, ethanol/acetone degradation, and immune/inflammatory responses like liver X receptor/retinoid X receptor regulation and complement system.

Conclusion: Our results suggested that CACYBP expression in HCC is associated with enhanced cell proliferation and cell cycle progression, as well as altered metabolic and immune responses, which contribute to the poorer clinical and survival outcomes of patients.

Abbreviations: CACYBP = calyculin-binding protein, DEG = differentially expressed gene, GO = gene ontology, GSEA = gene-set enrichment analysis, HCC = hepatocellular carcinoma, HR = hazard ratio, LHC = liver hepatocellular carcinoma (TCGA cohort), OS = overall survival, PPI = protein-protein interaction, PRISMA = Preferred Reporting Items for Systematic Reviews and Meta-Analyses, QUIPS = Quality in Prognosis Studies, TCGA = The Cancer Genome Atlas.

Keywords: breast invasive carcinoma, CACYBP, CX3CL1, hepatocellular carcinoma

RMO and JAY contributed equally to this work.

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All data generated or analyzed during this study are included in this published article [and its supplementary information files].

This article is review so ethics approval was not required for this study.

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1. Introduction

Cancer has become a significant worldwide public health issue, with an 18.1 million new cases and 9.6 million deaths in 2020 alone.^[1] Despite the widespread use of conventional treatments such as surgery, radiation, and chemotherapy, the popularity of targeted therapies and immunotherapy is growing.^[2,3] However, the lack of precise biomarkers that allow accurate prediction of prognosis and disease status in various types of tumors is a major barrier. The discovery of new biomarkers applicable to a wide range of cancers could significantly improve patient care.

The CACYBP gene encodes a highly conserved protein known as calcyclin-binding protein, which plays important roles in calcium signaling regulation, cell proliferation, and apoptosis.^[4,5] New research links CACYBP to the growth and progression of various cancer types, including colorectal cancer, non-small-cell lung cancer, and potentially hepatocellular carcinoma (HCC), among others.^[6,7] The implication of CACYBP in promoting cancer cell proliferation and invasion highlights its potential as a therapeutic target.^[6,7] However, many cancers have not undergone a thorough evaluation of CACYBP.

Even though there have been great improvements in treatment options such as liver transplantation and surgery, the prognosis of HCC, which is the most common type of liver cancer, is still poor, especially in advanced stages.^[8,9] Understanding the molecular mechanisms underlying HCC progression is key to

discovering new biomarkers and developing targeted therapy. Because CACYBP is involved in many cellular processes and is thought to play a role in the development of other types of cancer, it should be investigated in HCC. Despite CACYBP's widespread presence in tumors and its association with the progression of several cancer types,^[10–16] it has not extensively investigated its specific function and clinical importance in HCC. According to the new research, CACYBP levels may be associated with a poor outcome in HCC. One possible explanation is that CACYBP interacts with P27Kip1, a protein that regulates cell cycle.^[4] This discovery is especially fascinating given the well-established connection between cell cycle dysregulation and an unfavorable prognosis in individuals with HCC.^[17]

Furthermore, the restriction of numerous studies to limited sample sizes and certain forms of cancer adds complexity to the understanding of CACYBP's significance. In this study we conducted a systematically review, meta-analysis, and bioinformatic analysis to examine the prognostic importance of CACYBP expression in patients with HCC.

2. Methods

This systematic review and meta-analysis followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines.^[18] This study is a secondary analysis of published data; according to our institutional policy and Preferred

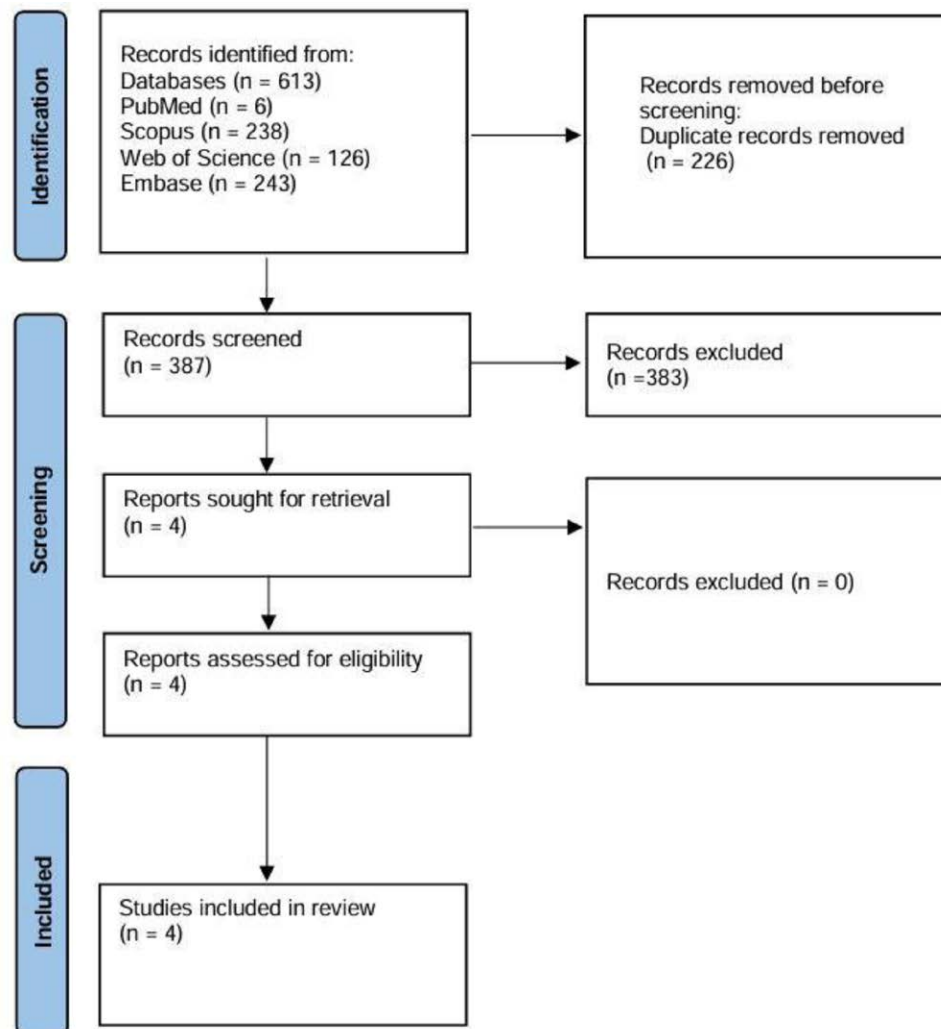


Figure 1. PRISMA flowchart. PRISMA = Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

Reporting Items for Systematic Reviews and Meta-Analyses guidance, ethics approval was waived.

2.1. Search strategy and eligibility criteria

PubMed, Scopus, Embase, and Web of Science databases were searched from inception to April 2024, using a combination of keywords related to hepatocellular carcinoma and CACYBP: (“CACYBP” OR “Calcyclin-binding protein”); (“Liver” OR “Hepatic” OR “Hepatocellular” OR “HCC” OR “Hepatoma”); (“Cancer” OR “Carcinoma” OR “Malignancy” OR “Neoplasm” OR “Tumor”); (“prognosis” OR “survival” OR “prognostic”). Moreover, we reviewed the reference lists of retrieved articles to complement the broad search.

The inclusion criteria for our study were as follows: 1. the searches were limited to studies that were conducted on human subjects with no language restrictions. 2. The focus of the study had to be on measuring the prognostic significance of CACYBP in hepatocellular carcinoma (HCC). 3. Only articles that were published in peer-reviewed international journals were included in our analysis. 4. Studies had to provide sufficient data for both qualitative and quantitative analysis. Studies that did not meet our criteria were excluded.

2.2. Study selection and data extraction

Two authors independently evaluated the titles and abstracts of all articles using the inclusion criteria. They also independently reviewed the full-text papers to determine if they met the criteria. The same researchers then independently extracted data from the studies using a data extraction Excel form. The final decision of whether to include each paper was made by consensus among all authors. Certain data were extracted from the included studies, including study characteristics and hazard ratios (HR). To ensure that no overlapping data were included and to maintain the integrity of the meta-analysis, the references and data for each included study were extensively cross-checked.

2.3. Critical appraisal tool and risk of bias assessment

Quality assessments of included studies were performed by 3 independent investigators. We used the Quality in Prognosis Studies (QUIPS) tool, recommended for quality assessment of

prognostic studies.^[19] The QUIPS tool evaluates various domains including study participation, study attrition, prognostic factor measurement, outcome measurement, confounding, and statistical analysis and reporting. Each domain was assessed for risk of bias, with an overall score reflecting the methodological quality of the study, classified as high, moderate, low, and no information studies. Studies achieving a score indicative of low risk of bias, typically above a predefined threshold, were considered to have a high-quality methodology. Any disagreements in the quality assessment were resolved by the third independent investigator.

2.4. Meta-analysis

Data from the primary studies were extracted and analyzed using Review Manager version 5.4.1. We used HR and corresponding 95% confidence interval to assess the association between CACYBP expression and overall survival (OS). An HR >1 indicated worse prognosis among patients exhibiting CACYBP overexpression. Heterogeneity was evaluated using the inconsistency (I^2) and Chi-squared (X^2) tests. A measure of heterogeneity, I^2 , was used to assess the variation in study outcomes. A value of $I^2 > 50\%$ was considered to indicate substantial heterogeneity, while a value >90% was considered to indicate major heterogeneity. A P value <.05 was considered statistically significant. Due to the limited number of studies included in the analysis, we were unable to conduct funnel plot or publication bias analyses as suggested by Cochrane guidelines.

2.5. Transcriptomic exploration

2.5.1. The Cancer Genome Atlas (TCGA) analysis and validation. We conducted several analyses utilizing liver hepatocellular carcinoma (LIHC) cohort. We used gene expression profiling interactive analysis: a web server for cancer and normal gene expression profiling and interactive analyses^[20] for differential expression (limma) analysis of CACYBP gene between tumor and normal samples ($|\log_2FC| > 1$ and $q < 0.01$), and for expression boxplots. We performed Kaplan–Meier survival analysis using TCGA data to validate the prognostic implications of CACYBP expression identified in the meta-analysis. Tumor Immune Estimation Resource: A web server for comprehensive analysis of tumor-infiltrating immune cells,^[21–23]

Table 1
Characteristics of studies.

Study	Year	Number of participants	Aim	Conclusion
Hu et al.	2020	378	The clinical relevance and prognostic significance of IRGs in Liver hepatocellular carcinoma must be explored.	Our findings screen several IRGs with clinical significance, reveal the drivers of immune repertoire, and illustrate the importance of a personalized, IRG-based immune signature in Liver hepatocellular carcinoma recognition, surveillance, and prognosis prediction.
Mo et al.	2020	752	This research aimed to explore the potential correlation between the mTORC1 signal pathway and Liver hepatocellular carcinoma and establish the related gene signature.	The mTORC1-associated gene signature established and validated in our research could be used as a potential prognostic factor in Liver hepatocellular carcinoma.
Peng et al.	2021	372	we aimed to investigate whether immune-related genes could be indicators for the prognosis of HCC patients and TME cell infiltration characterization as well as responses to immunotherapy.	The mutation analysis demonstrated a significant difference in the tumor mutation burden (TMB) between the high- and low-risk groups, partially explaining this signature's predictive value.
Shi et al.	2023	600	We aimed to establish a risk model based on Tex related genes for HCC prognosis	Our study demonstrated that Tex-related genes might provide accurate prediction for HCC patients in clinical decision-making, prognostic assessment, and immunotherapy. In addition, targeting the hub genes or transcription factors may help to reverse T cell function and enhance the effect of tumor immunotherapy.

HCC = hepatocellular carcinoma, IRGs = immune-related genes, mTORC1 = mechanistic target of rapamycin complex 1, Tex = T cell exhaustion, TMB = tumor mutation burden, TME = tumor microenvironment.

was employed for multivariate cox-proportional hazards model considering CACYBP expression, and for generating Kaplan-Meier plots.

2.5.2. Differential expression, protein-protein interaction (PPI) network, and functional enrichment analysis. We used “DESeq2” package^[24] on R to identify differentially expressed genes (DEGs) ($|\log_2FC| > 1$ and $q < 0.05$) between high and low CACYBP expressors. Patients with expression $> Q3$ (third quartile) were considered high-expressors, while those with CACYBP expression $< Q1$ (first quartile) were considered low-expressors. Gene-set enrichment analysis (GSEA) on up- and down-regulated genes in high-expressors (relative to low-expressors) was performed using “TCGAbiolinks” package^[25–27] on R, integrating 4 gene ontology (GO) databases, including GO: biological process, GO: molecular function, GO: cellular component, and pathways. Finally, we constructed a PPI network using the Search Tool for the Retrieval of Interacting Genes/Proteins database^[28] to shed light on possible connections and interactions involving CACYBP.

3. Results

3.1. Search results

Our search strategy yielded 613 studies from Medline (via PubMed), Web of Science, Scopus, and Embase. Following the removing of duplicate records, 387 studies underwent screening based on title and abstract, resulting in the exclusion of 4 studies. The remaining 4 full texts were independently reviewed by the same authors. Of these, 4 were deemed eligible for inclusion.^[29–32] Meta-analysis was conducted on 4 of these studies. The search and selection process are illustrated in Fig. 1.

3.2. Summary of the included studies

Our study included a total of 2102 participants from 4 studies.^[29–32] Two studies had been published in 2020, one in 2021 and one in 2023. There is no study whose main objective was to study CACYBP expression alone in HCC (Table 1).

3.3. Quality assessment

We performed a quality assessment for the 4 cohort studies using the QUIPS tool. All studies were judged to have a low risk of bias, indicating good methodological quality. However, moderate risk of bias was noted in the domain of outcome measurement (D4) for 3 studies (Hu et al^[29], Mo et al^[30], and Peng et al^[31]), and in the domain of confounding (D5) for one study (Peng et al^[31]). The details of the domain-specific assessments are summarized in Fig. 2.

3.4. Meta-analysis

Four studies were included in the analysis.^[29–32] The pooled HR for OS demonstrate that there is a significant association between high CACYBP expression and poor prognosis (HR = 1.08; 95% confidence interval: 1.05–1.10, $I^2 = 0\%$, P value $< .00001$) (Fig. 3).

3.5. TCGA validation

To validate our result, we retrieved CACYBP expression data and clinical data from the TCGA dataset. CACYBP was increased in hepatocellular carcinoma (HCC) determined using a $\log_2FC$ cutoff of 1 and a P -value cutoff of (.01) (Fig. 4A). A total of 371 patients with HCC were included in the survival analysis from

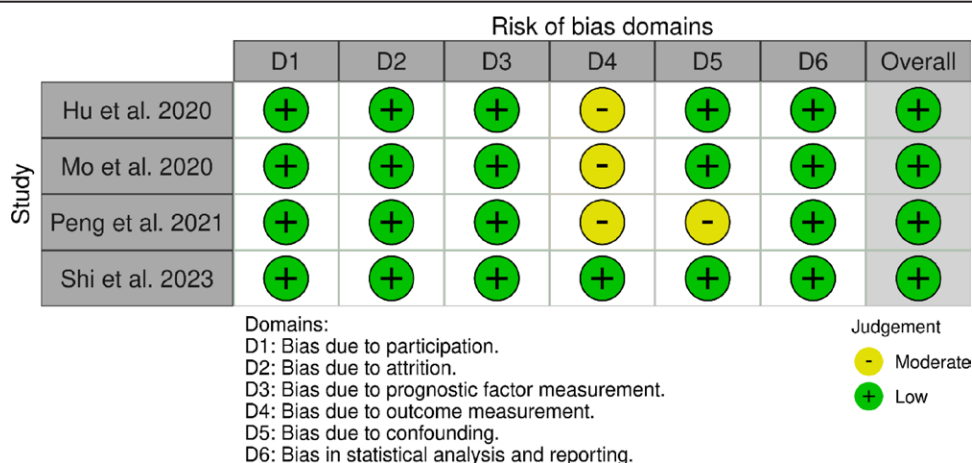


Figure 2. Quality assessment (QUIPS). QUIPS = quality in prognosis studies.

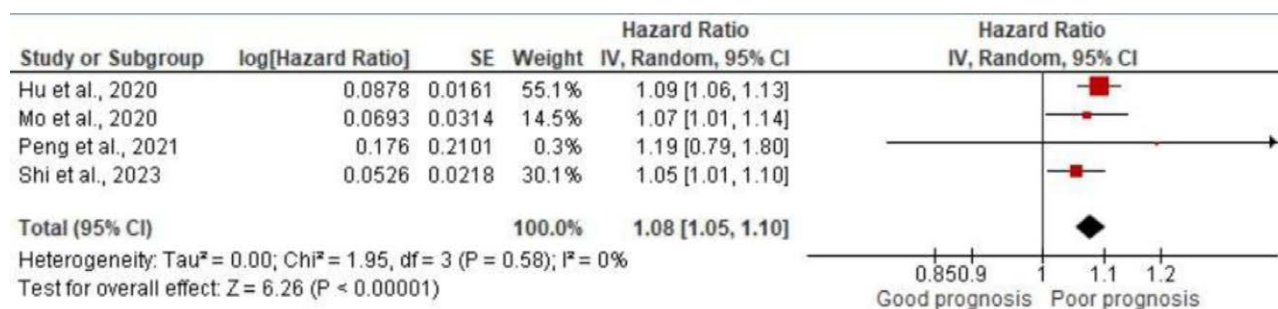


Figure 3. Forest plots for meta-analysis on OS. OS = overall survival.

the TCGA database. The results showed that higher CACYBP expression was correlated with poor prognosis, as indicated by univariate HR (HR: 1.24, P -value: .04). However, the association between CACYBP expression and prognosis seems to be time dependent (Fig. 4B).

The multivariate cox-proportional hazards model adjusted by CACYBP, age, gender, race, stage, and tumor purity, showed significant associations with CACYBP expression, age, stage

III, and stage IV, further confirming the robustness of CACYBP prognostic value (Table 2).

3.6. Transcriptomic exploration

We constructed a Search Tool for the Retrieval of Interacting Genes/Proteins PPI network. The network revealed interactions between CACYBP and RNF41, S100A6, SKP1, SIAH1, GSK3A, CTNNB1, GSK3B, APC, AXIN1, HSP90AA1, and several other proteins (Fig. 5).

Two thousand nine hundred ten genes differentially expressed (1846 up-regulated and 1047 down-regulated) between high and low CACYBP expression groups, significantly impacting HCC biology. GSEA revealed upregulated genes in CACYBP high-expression groups are associated with nuclear division, mitosis, and various phases of the cell cycle, especially the M and S phases, suggesting a role of CACYBP in promoting cell proliferation and division processes. Upregulated genes in patients with high CACYBP expression are also linked to the microtubule cytoskeleton and chromosome condensation, hinting at CACYBP's involvement in organizing the cytoskeleton and managing chromosomal dynamics during cell division. Pathways enriched among upregulated genes in high CACYBP expressors include those related to the mitotic roles of Polo-like kinase, cell cycle control, and DNA damage response, implying a significant impact of CACYBP on cell cycle regulation and DNA repair mechanisms.

Conversely, GSEA showed downregulated genes in high CACYBP expressors related to oxidation-reduction, steroid and cholesterol metabolism, and responses to organic substances and hormones. This may indicate a metabolic shift or altered cellular responses associated with higher CACYBP levels, in addition to extracellular region and functions such as oxygen and electron carrier activities, pointing to changes in cellular interactions and transport activities.

Shared pathways affected by up- and down-regulated genes encompassed liver X receptor/retinoid X receptor activation, acute phase response signaling, and the complement system, highlighting the broad impact of CACYBP on different cellular signaling pathways and immune responses in LIHC. Overall, these findings suggest that high CACYBP expression in LIHC is associated with enhanced cell proliferation and cell cycle progression, as well as altered metabolic and immune responses, which contribute to the poorer clinical outcome of these patients (Fig. 6A, B, and C).

4. Discussion and conclusion

This study highlighted the significant relationship between CACYBP expression and HCC prognosis. Our results indicated that upregulation of CACYBP in HCC patients correlated with poor prognosis. Additionally, a significant number of DEGs were identified between the high and low CACYBP expression groups. However, a larger number of studies with longer follow-up durations are needed to further validate these outcomes.

Liver cancer, primarily HCC, is the second leading cause of cancer-related deaths globally.^[33,34] While there has been limited progress in developing clinically effective biomarkers for early surveillance and diagnosis of HCC, biomarkers may still play a role in prognosis.^[35] Recent research in HCC has unveiled multiple biomarkers that hold considerable prognostic value, including different hub genes such as cyclin B1, cyclin B2 and ZW10 interacting kinetochore protein,^[36,37] PDZ binding kinase.^[38] Moreover, CACYBP expression has been found to be upregulated in HCC patients, and prior research has highlighted CACYBP as a significant prognostic marker in HCC, indicating worse outcomes associated with its upregulation.^[4,31]

CACYBP has also been found to be a key prognostic biomarker for multiple human cancers, where elevated expression

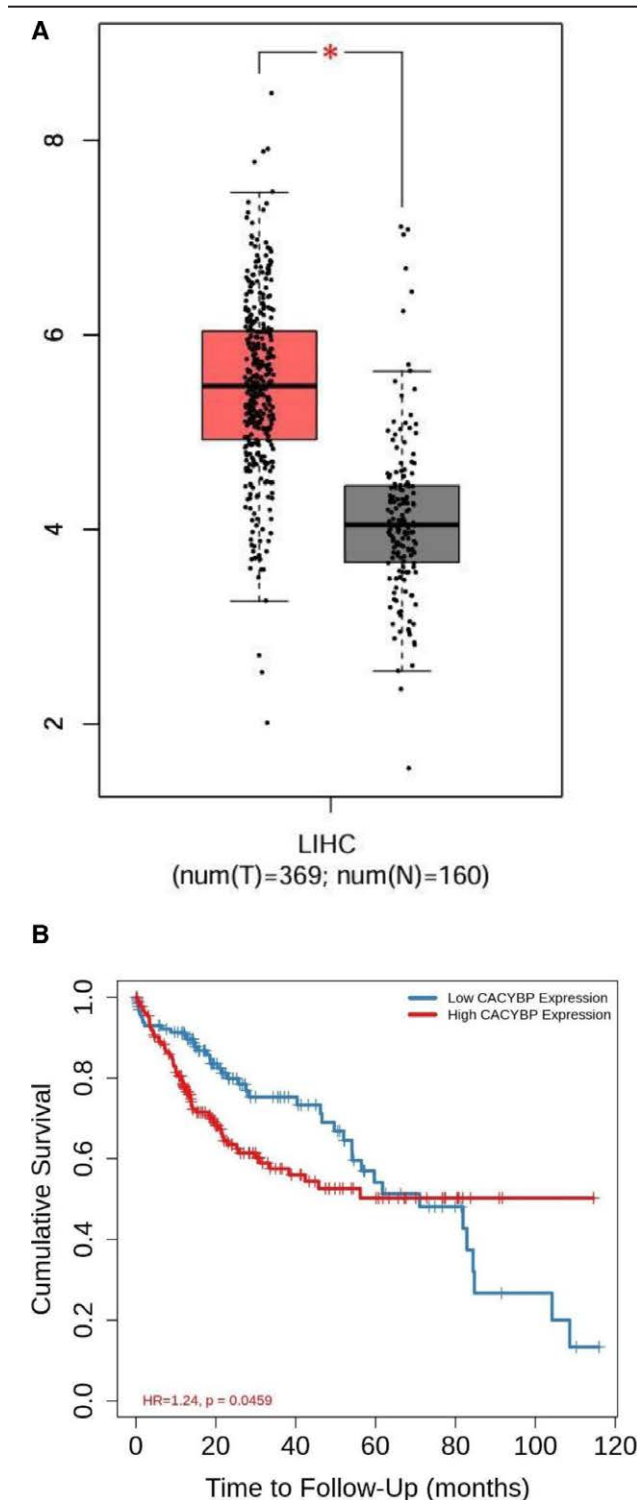


Figure 4. (A) CACYBP expression in HCC. (B) Kaplan-Meier (KM) of OS based on TCGA data. CACYBP = calcyclin-binding protein, OS = overall survival.

Table 2
Multivariate analysis results based on 371 patients recruited from TCGA cohort.

Variables	HR	SE	95% CI	Z-score	P-value
CACYBP	1.703	0.143	1.287, 2.253	3.723	<.001
Age	1.016	0.008	1.000, 1.032	1.910	.056
Gender (male)	0.947	0.227	0.606, 1.478	-0.241	.810
Race (black)	2.479	0.493	0.943, 6.522	1.840	.066
Race (white)	1.243	0.243	0.772, 1.999	0.896	.370
Stage II	1.138	0.267	0.675, 1.919	0.484	.628
Stage III	2.400	0.235	1.515, 3.802	3.728	<.001
Stage IV	4.693	0.619	1.396, 15.781	2.499	.012
Purity	1.328	0.467	0.532, 3.314	0.607	.544

CACYBP = calcyclin-binding protein, CI = confidence interval, CPH = cox proportional hazards, HR = hazard ratio, SE = standard error, TCGA = The Cancer Genome Atlas.

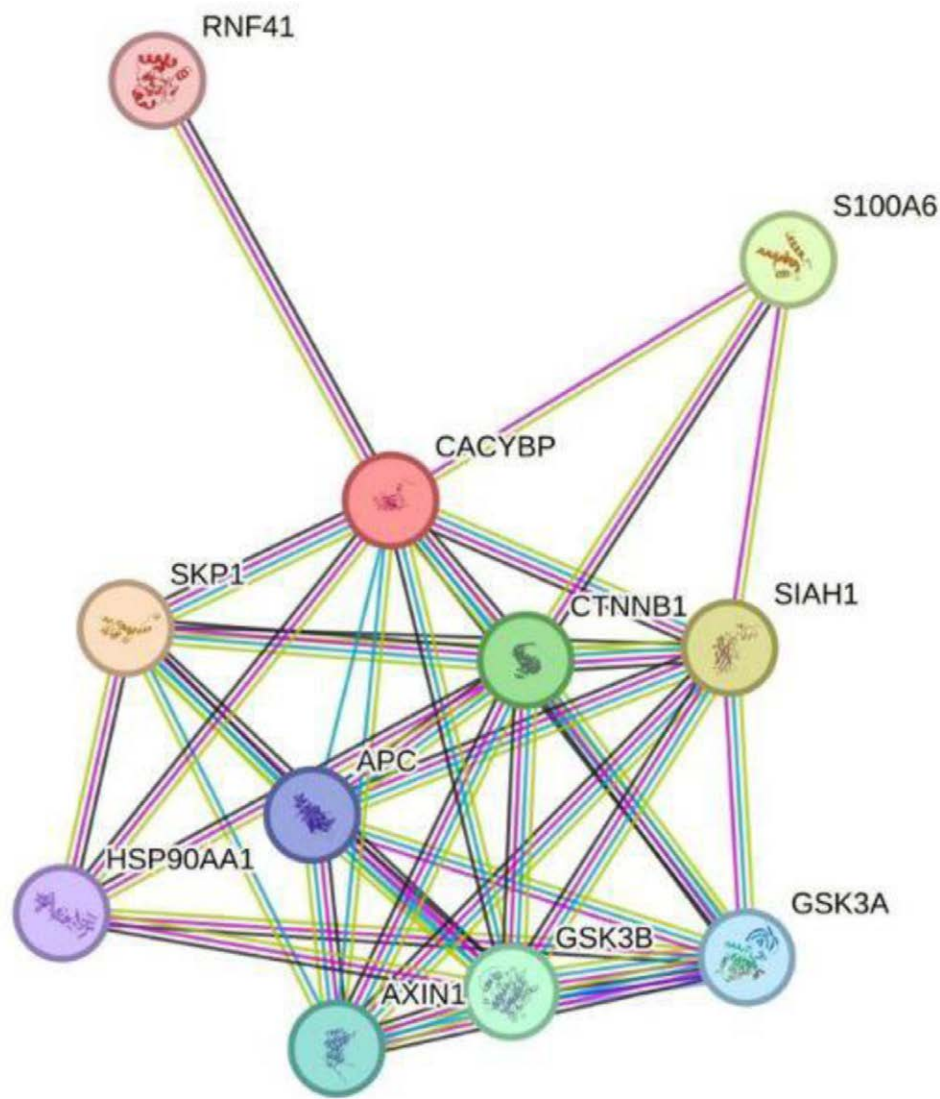


Figure 5. PPI network for CACYBP using STRING. CACYBP = calcyclin-binding protein, PPI = protein–protein interaction, STRING = Search Tool for the Retrieval of Interacting Genes/Proteins.

of this gene translated into more unfavorable outcomes in bladder urothelial carcinoma, breast invasive carcinoma, lung adenocarcinoma, glioblastoma, and osteosarcoma.^[14,39–42] Although, recent studies have shown a more complex relationship between CACYBP and cancer, with heightened expression playing a protective role in kidney renal clear cell carcinoma.^[43]

CACYBP participates in a broad array of biological processes by binding with several target proteins interacting with cytoskeletal proteins like tubulin, actin, and tropomyosin, and modulating cytoskeleton dynamics.^[44,45] It also facilitates the ubiquitination and degradation of non-phosphorylated β -catenin, while also acting as a phosphatase to inhibit the

mitogen-activated protein kinase signaling cascade through binding to and dephosphorylating extracellular signal-regulated kinase 1/2 and P38.^[46,47] Previous studies found that high levels of CACYBP in HCC are linked to reduced response to anti-PD-1 treatment by interacting with myeloid differentiation primary response 88, preventing its degradation, and leading to increased expression of chemokine (C-X3-C motif) ligand 1. This results in immune evasion and reduced response to anti-PD-1 therapy.^[48] Moreover, CACYBP exerts its tumor-promoting effect by releasing rapid cell cycle progression through promoting the Ser10 phosphorylation and subsequent cytoplasmic sequestration of P27Kip1, while its depletion inhibits HCC cell proliferation and growth. Significant associations between elevated CACYBP expression and increased alpha-fetoprotein levels, elevated aspartate aminotransferase levels, decreased albumin levels, as well as heightened occurrences of mortality and recurrence were also previously found.^[4]

Our findings highlighted the elevated expression of CACYBP in HCC tumor samples compared to normal ones, which was consistent with the existing literature. CACYBP in HCC patients had a multivariate hazard ratio of 1.7 for OS, indicating poorer prognosis with higher expression of this gene. Additionally, we identified a total of 2910 DEGs, with 1047 being downregulated and 1846 upregulated in the high CACYBP group compared to the low CACYBP group. Shared pathways affected by up- and down-regulated genes encompassed liver X receptor/retinoid X receptor activation, acute phase response signaling, and the complement system, highlighting the broad impact of CACYBP on different cellular signaling pathways and immune responses in HCC.

Our results indicated that elevated CACYBP expression was associated with the upregulation of gene sets linked to nuclear

division, mitosis, and various phases of the cell cycle, particularly the M and S phases. This suggests CACYBP's role in promoting cell proliferation and division processes. The upregulated genes in patients with high CACYBP expression were also found to be associated with the microtubule cytoskeleton and chromosome condensation, indicating CACYBP's involvement in organizing the cytoskeleton and managing chromosomal dynamics during cell division. Pathways enriched among upregulated genes in high CACYBP expressers included those related to the mitotic roles of cell cycle control, polo-like kinase, and DNA damage response, suggesting a significant impact of CACYBP on DNA repair mechanisms and cell cycle regulation.

On the other hand, high CACYBP expression led to relative downregulation of genes involved in oxidation-reduction, steroid and cholesterol metabolism, and responses to organic substances and hormones in comparison with low expressers. This may indicate a metabolic shift or altered cellular responses associated with higher CACYBP levels. In addition, downregulated gene sets also included those related to the extracellular region and functions such as oxygen and electron carrier activities, indicating changes in cellular interactions and transport activities. Finally, the PPI network analysis revealed significant interactions between CACYBP and several key proteins, including RNF41, S100A6, SKP1, SIAH1, and components of the GSK3 and CTNNB1 complexes, suggesting a carcinogenic role.

This study has several limitations, including the variability in measurement techniques for CACYBP expression, with the meta-analysis depending on cohorts from overlapping sources and the bioinformatics analysis using messenger RNA expression data, mainly. Additionally, the meta-analysis was based on only 4 studies, which limits the strength and generalizability of our findings. Heterogeneity in patient demographics, treatment regimens, and clinical stages across the included studies may

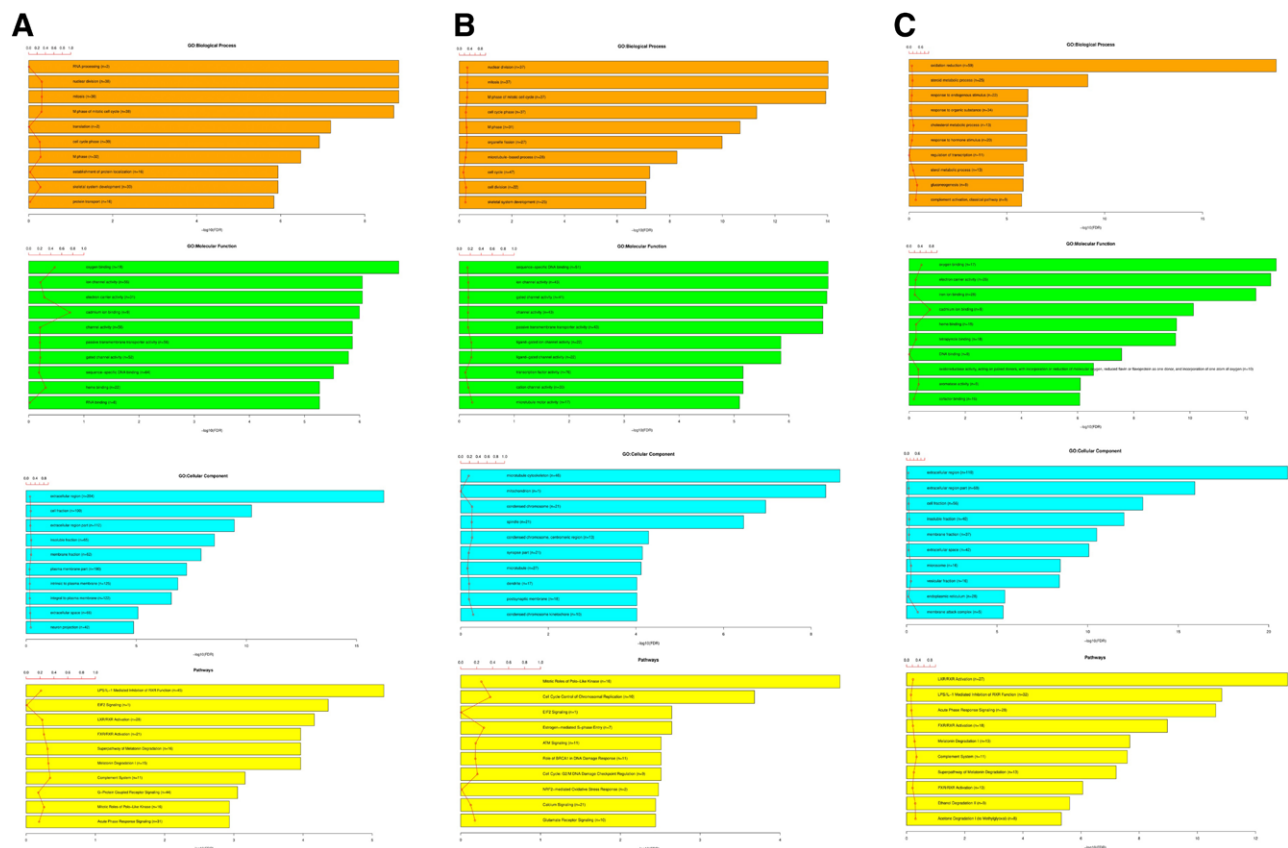


Figure 6. (A) Gene set enrichment for all CACYBP. (B) Gene set enrichment for upregulated CACYBP. (C) Gene set enrichment for downregulated CACYBP. CACYBP = calyculin-binding protein.

introduce bias. The inability to assess publication bias due to the limited number of studies further affects the validity of the results. Future research should focus on standardizing measurement techniques, including larger and more diverse cohorts, conducting long-term follow-up studies, and performing functional studies to explore the mechanistic role of CACYBP in HCC.

Overall, these findings suggest that high CACYBP expression in HCC is associated with enhanced cell proliferation and cell cycle progression, as well as altered metabolic and immune responses, which contribute to the poorer clinical outcome of these patients. These attributes underscore the possibility of CACYBP's potential as a therapeutic target for cancer as well as an important predictor of cancer outcomes.

Author contributions

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